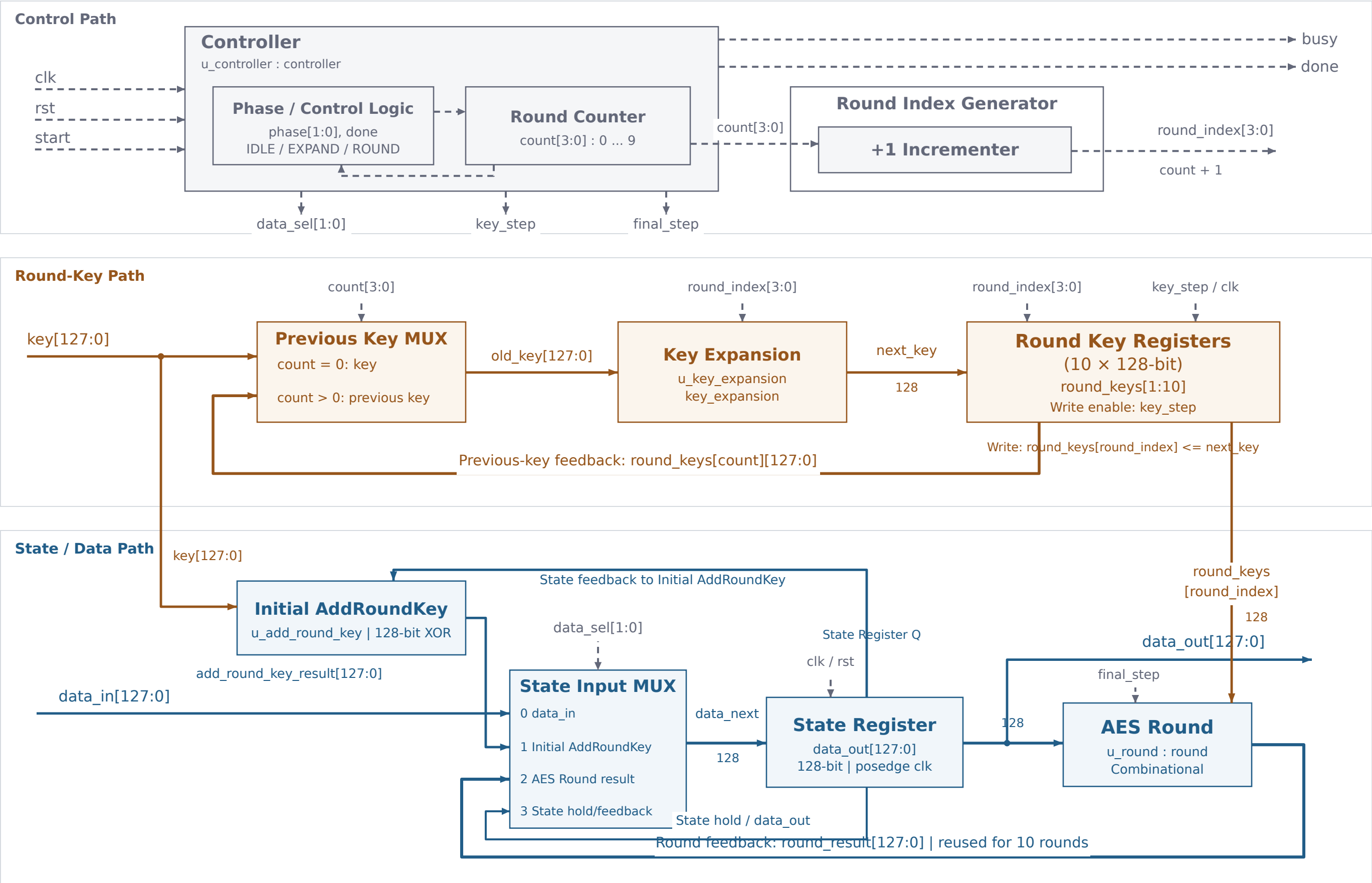


Top-Level Architecture

aes128_iterative_core | 128-bit iterative encryption | 10 key-expansion clocks + 10 round clocks

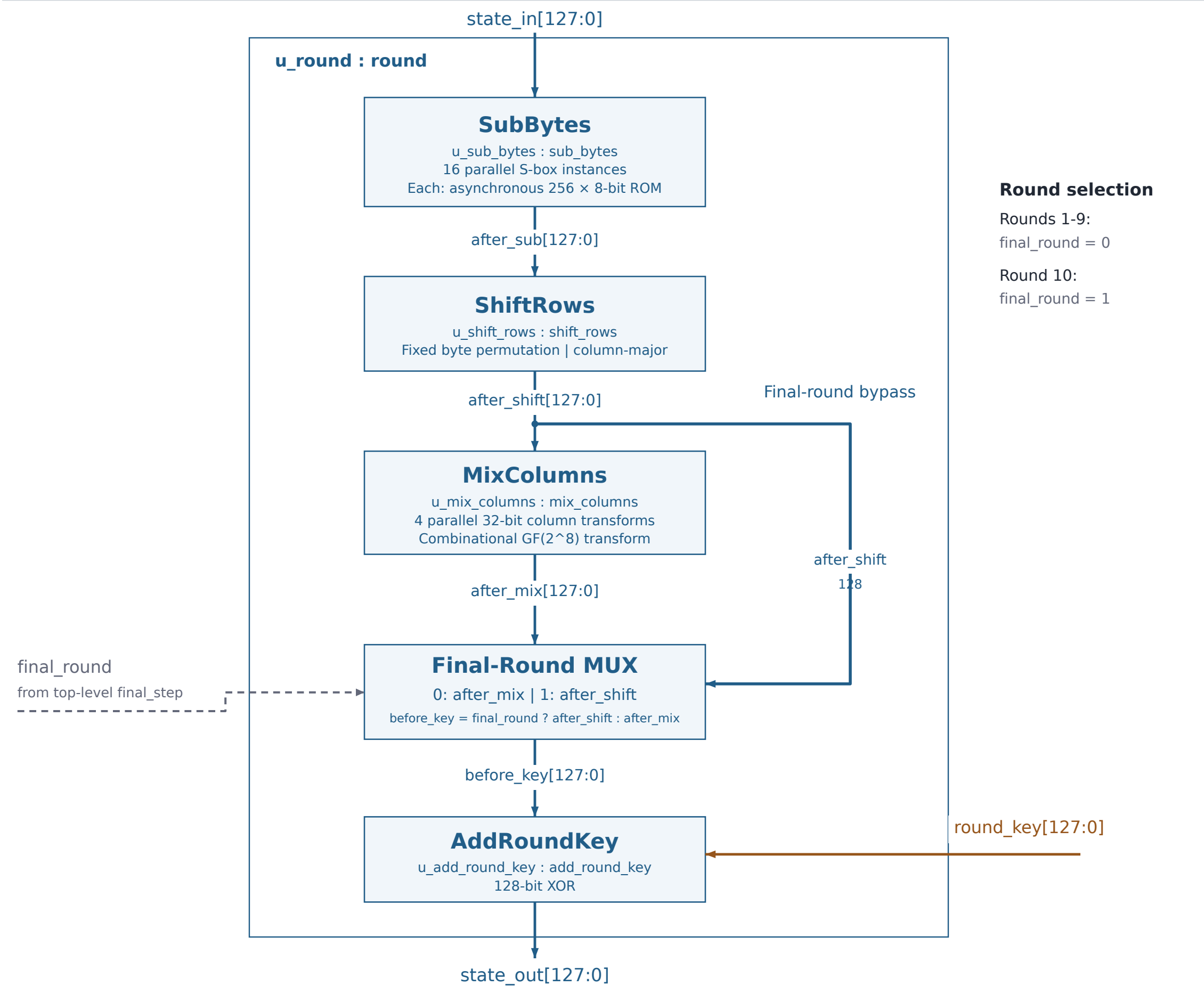


State / Data Round Key Control / address / clock

rst: synchronous, active high; resets Controller and State Register only. Round Key Registers have no reset.

Matching control labels denote the same net.

round | one combinational datapath reused across all 10 encryption rounds



No internal state registers or pipeline stages. MixColumns is bypassed by selection; no clock gating is present.

Key Expansion Architecture

FIG. 03

key_expansion | one combinational 128-bit round-key update | round_index = 1 ... 10

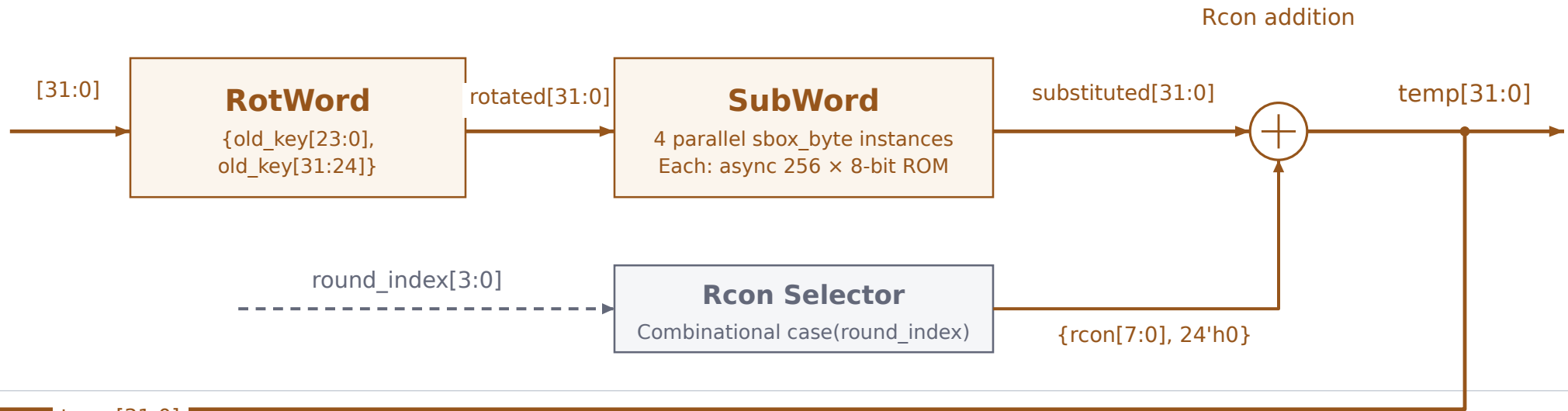
Key schedule transform

old_key[127:0]

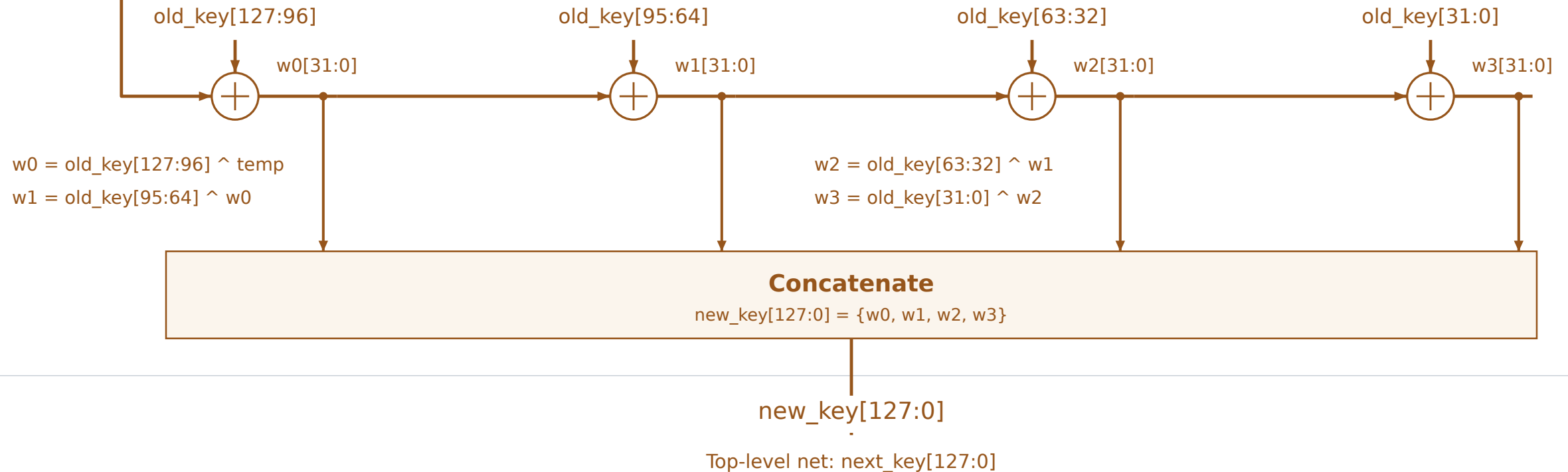
Word split (wiring)

[127:96] [95:64]

[63:32] [31:0]

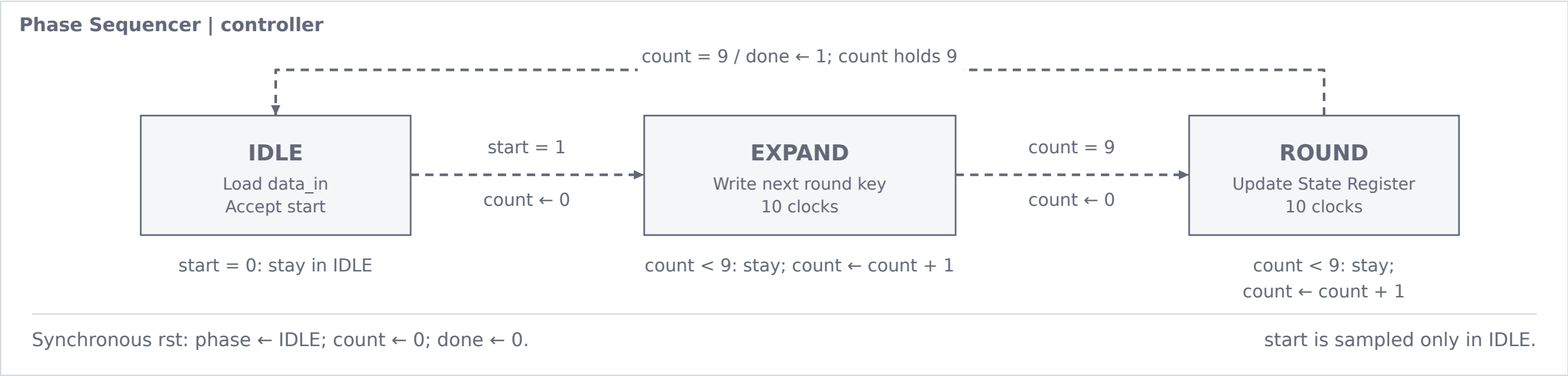


32-bit XOR chain



No key registers inside this module. Rcon is selected from constants; all XOR-chain values are combinational.

Three-phase controller with a shared round counter and a one-clock done pulse



Combinational forward transforms and byte organization implemented by the RTL

AES State | column-major byte ordering

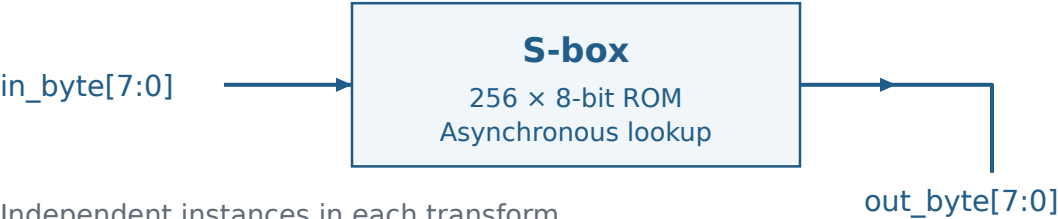
128-bit state

b0 = state[127:120]
b15 = state[7:0]

	col 0	col 1	col 2	col 3
r0	b0	b4	b8	b12
r1	b1	b5	b9	b13
r2	b2	b6	b10	b14
r3	b3	b7	b11	b15

Byte index = 4 × column + row
state = {b0, b1, ..., b15}

S-box Lookup and RTL Replication



Independent instances in each transform



ROM image: mem/sbox.mem; no registered S-box output.

ShiftRows

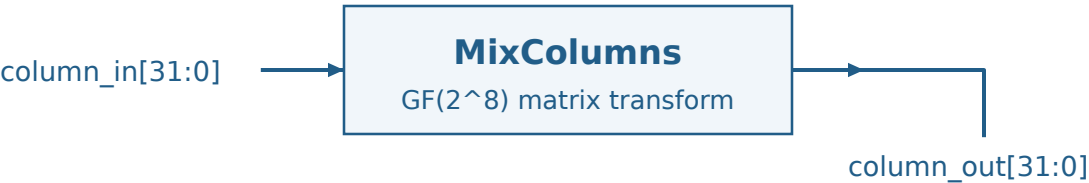
Fixed byte permutation; row r rotates left by r bytes.



out[r,c] = in[r,(c+r) mod 4]

MixColumns

Four independent 32-bit column transforms

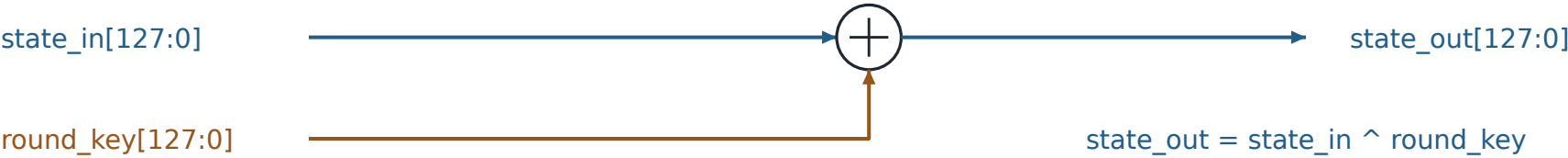


M =

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

Applied to each column
Reduction polynomial: 0x11B
No pipeline registers

AddRoundKey | 128-bit XOR



Two RTL instances: Initial AddRoundKey at the top level, and AddRoundKey inside AES Round.